

RAVEN PROMPT STRATEGY

INITIAL PROMPT:

CREATE A PARAMETRIC, ASYMMETRICAL HELIX TOWER WITH A HEXAGONAL FACADE.

EVALUATION – I STARTED WITH A SIMPLE PROMPT STRATEGY TO BUILD AN INITIAL GEOMETRY AND SCRIPT BASE. I DID NOT WANT TO WRITE AN OVERLY COMPLEX OR CONSTRAINING INITIAL PROMPT THAT MAY RESULT IN ERRORS. TO MY SURPRISE, RAVEN BUILT-IN SEVERAL ADDITIONAL PARAMETER CONTROLS, SEPARATING THE *U & V DIVISIONS* WITHOUT REQUEST, *BASE RADIUS* TAKING LIBERTY TO GENERATE A CIRCULAR TOWER, *ECCENTRICITY* IN RESPONSE TO “ASYMMETRICAL”, *FLOOR HEIGHT & FLOOR COUNT* IN LEU OF A SINGLE HEIGHT CONTROL, AND *TWIST ANGLE* PICKING UP ON “HELIX”.

FOLLOW UP PROMPTS:

REVISE THE FLOOR COUNT SLIDE TO UTILIZE WHOLE NUMBERS ONLY.

ADD PIPE GEOMETRY TO THE HEXAGONAL FACADE GRID WITH A SLIDER TO CONTROL PIPE DIAMETER.

EVALUATION – WITH THE INITIAL GEOMETRY AND SCRIPT IN PLACE I MOVED TO MORE SPECIFIC PROMPTS TARGETING INDIVIDUAL PARAMETER CONTROLS. RAVEN TOOK THE LIBERTY OF ADDING A CUSTOM PREVIEW AND SWATCH WIDGET WHICH PROVED HELPFUL IN ADJUSTING THE DISPLAY. DESPITE THIS, THE HEXAGONAL FACADE WAS DIFFICULT TO READ VISUALLY EVEN AFTER ADJUSTING SEVERAL OF THE SLIDE VALUES.

FOLLOW UP PROMPT:

CHANGE HEXAGONAL FACADE GRID TO A DIAGRID.

EVALUATION – THE CHANGE TO A DIAGRID WAS IMPLEMENTED WITHOUT INCIDENT AND RESULTED IN A CLEARER GEOMETRY. AS I AM LARGELY UNFAMILAR WITH GRASSHOPPER, I QUERYED RAVEN ON HOW ADJUST ADDITIONAL VISUAL SETTINGS, EITHER BY HIDING ELEMENT IN THE PREVIEW OR ADDING AN ADDITIONAL CUSTOM PREVIEW AND SWATCH WIDGET TO THE LOST SURFACE.

FOLLOW UP PROMPT:

ORGANIZE THE SCRIPT INTO LABELED GROUPS

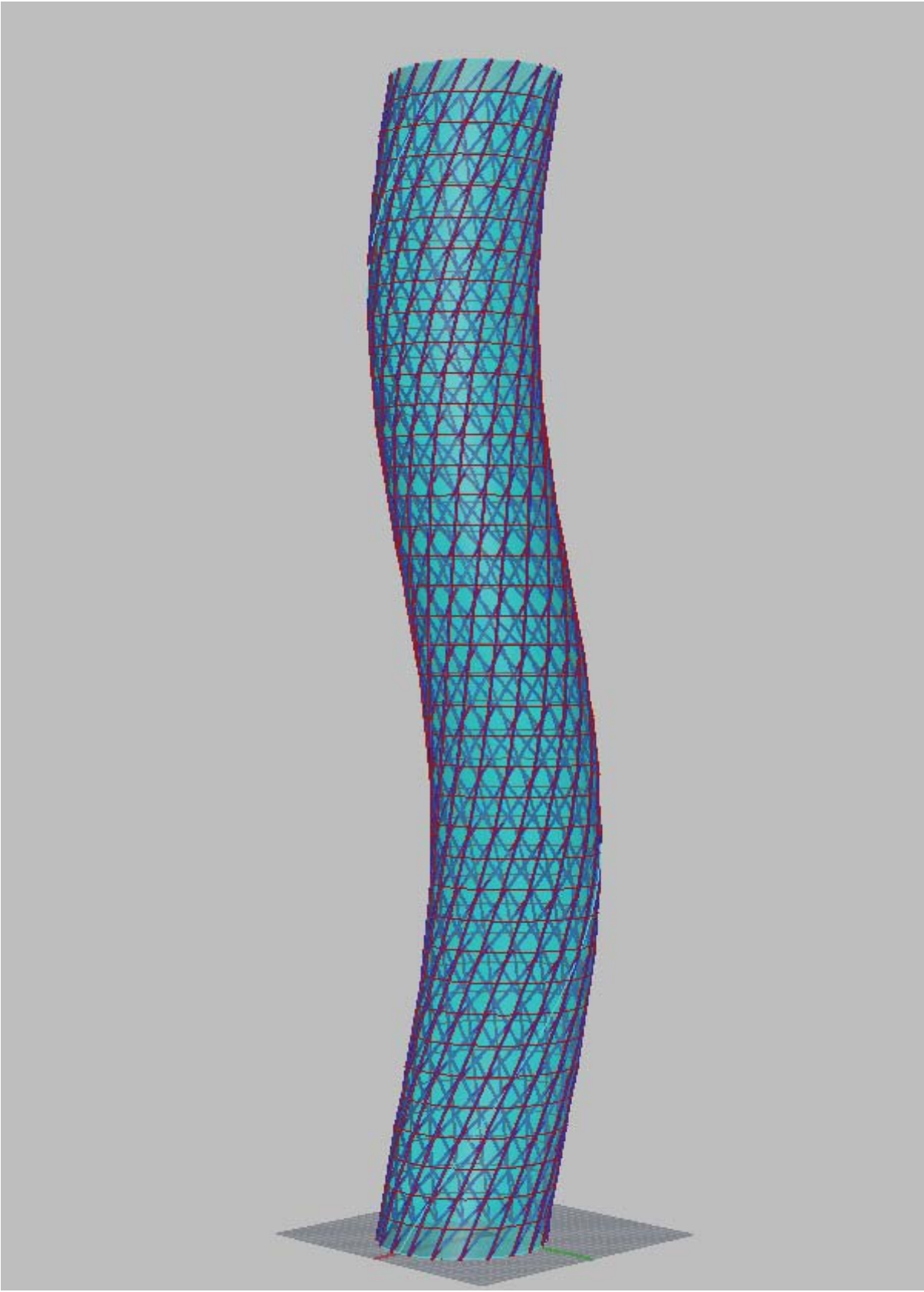
EVALUATION – THE SWATCH WIDGETS WERE GROUPED WITH THE OTHER PARAMETER CONTROLS, BUT THIS LEAD TO CROSSED WIRES OR OVERLAPPING GROUPS. THE SWATCHES WERE MANUALLY REMOVED FROM THE *PARAMETER GROUP*.

TOWER & SCRIPTS 1.0 & 1.1 ARE DEVIATIONS AT THIS STAGE ADJUSTING VARIOUS PARAMETER CONTROLS

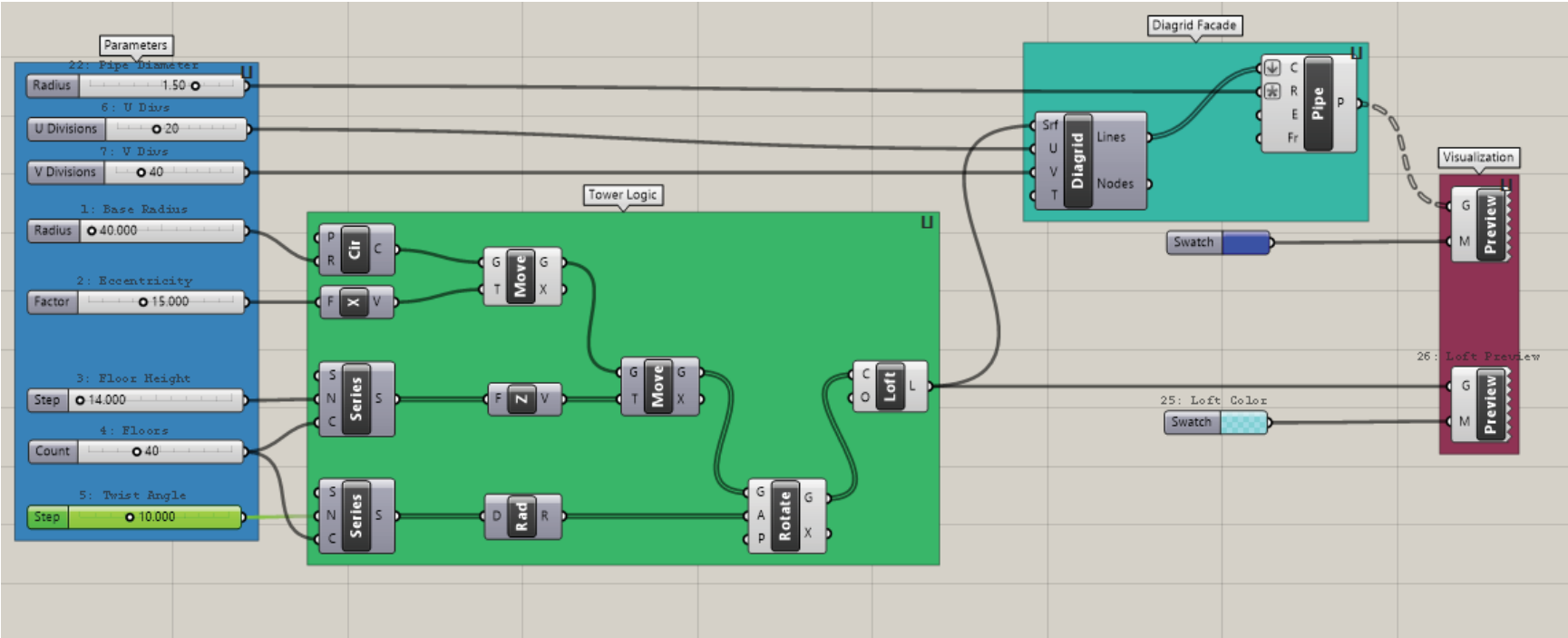
FINAL PROMPT:

MODIFY THE SCRIPT TO UTILIZE A TRIANGLE PLANE INSTEAD OF A CIRCLE BASE PLANE AND ADD A SCALAR CONTROL FOR EACH TOWER FLOOR

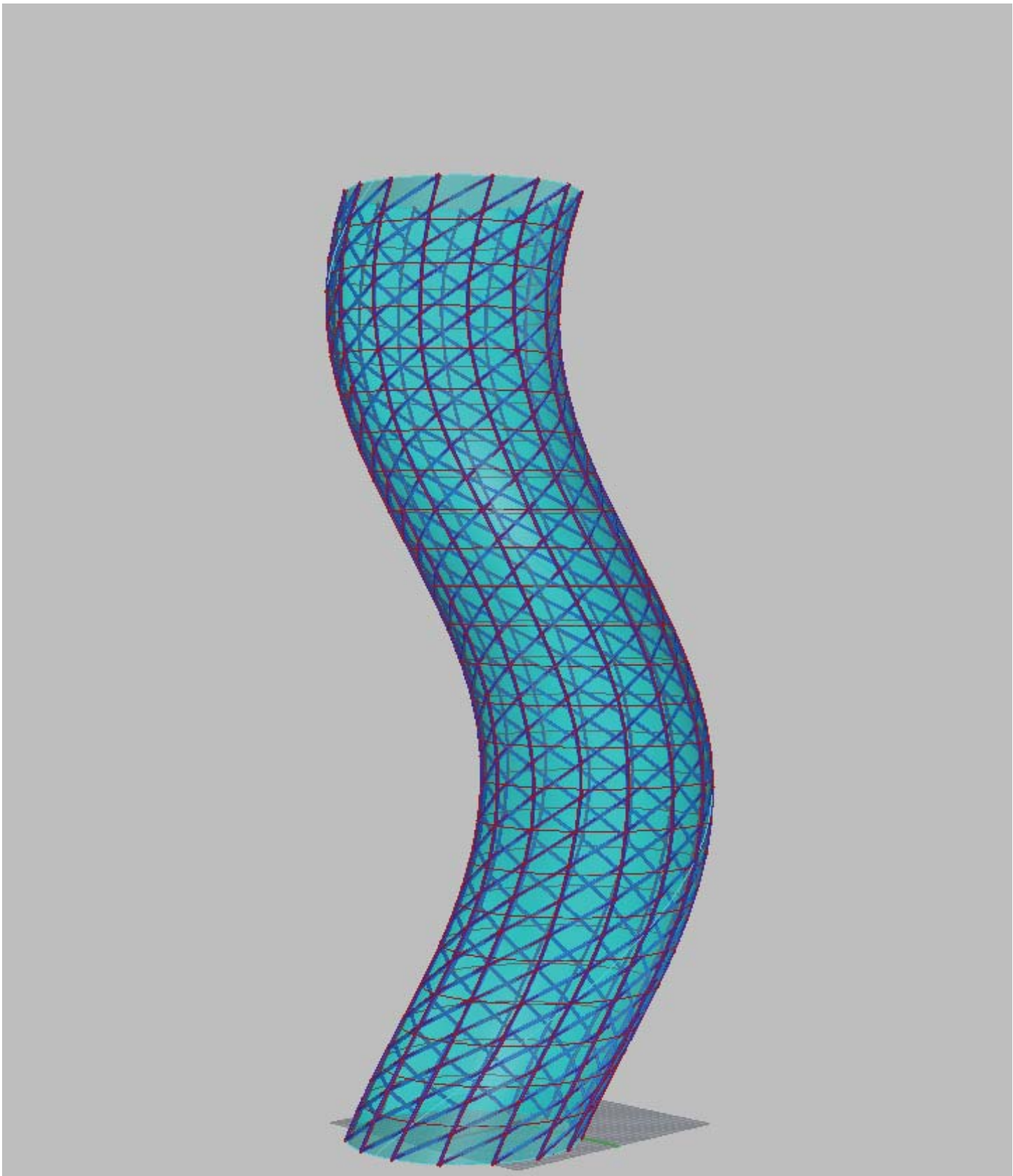
EVALUATION – DESPITE THE *ECCENTRICITY* CONTROL THE TOWER GEOMETRY STILL FELT STATIC AND NOT AS ASYMMETRICAL AS DESIRED. A TRIANGLE BASE PLANE WITH A TAPERING GEOMETRY PROVIDED A MORE SATISFYING RESULT.



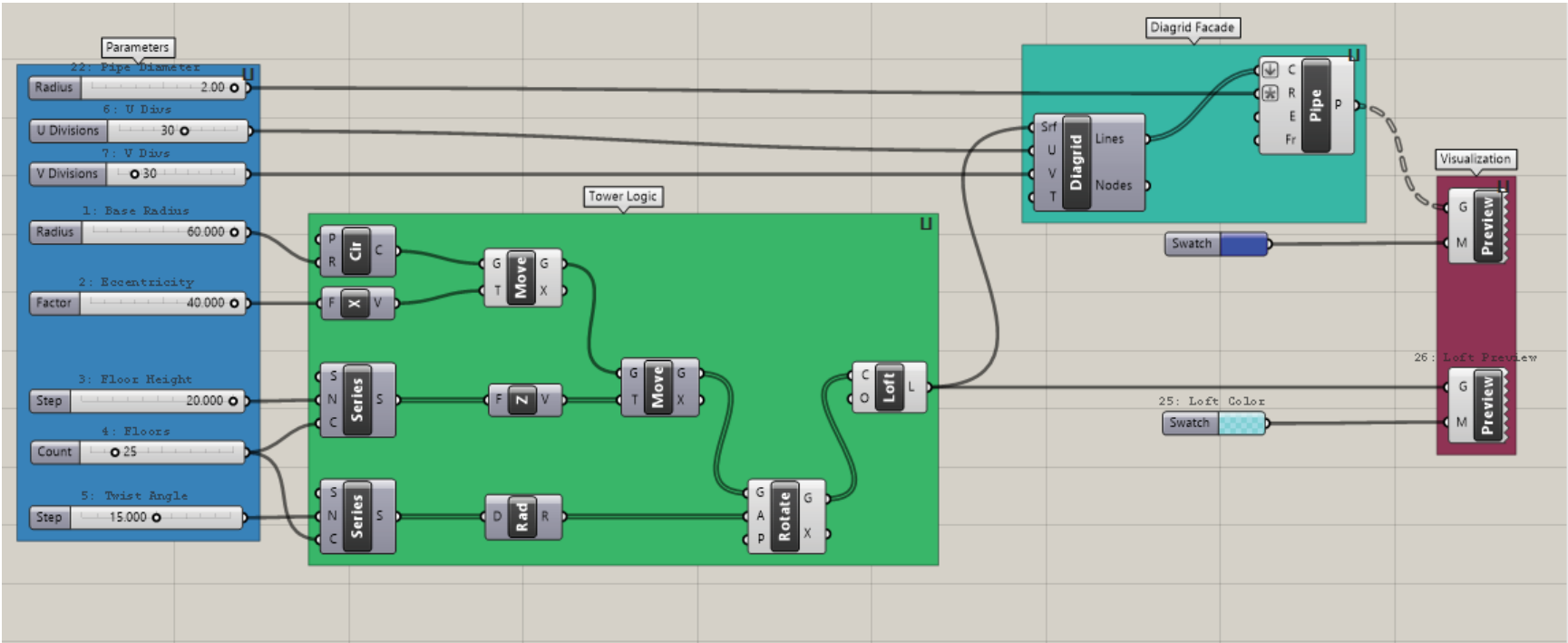
TOWER 1.0



SCRIPT 1.0

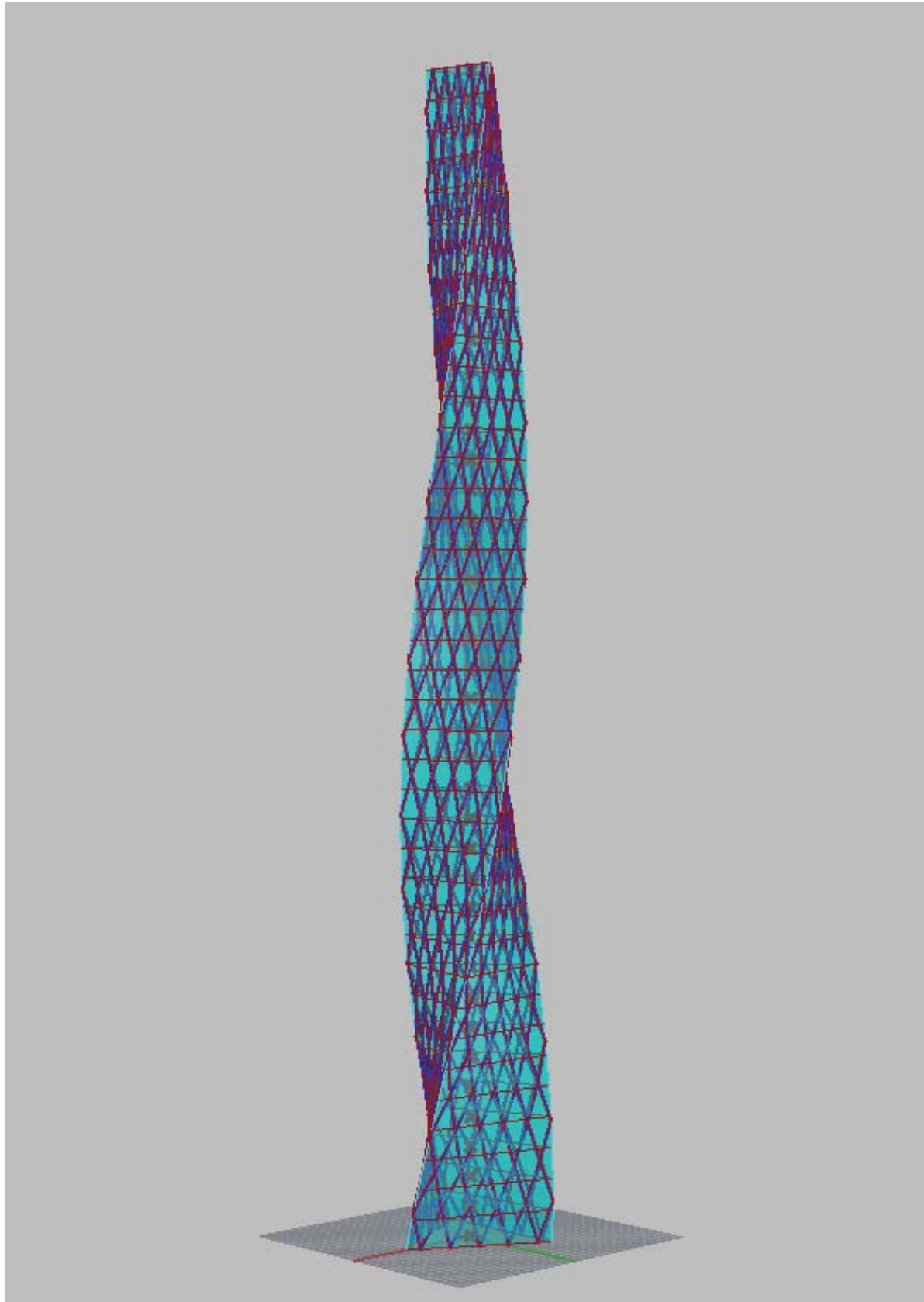


TOWER 1.1

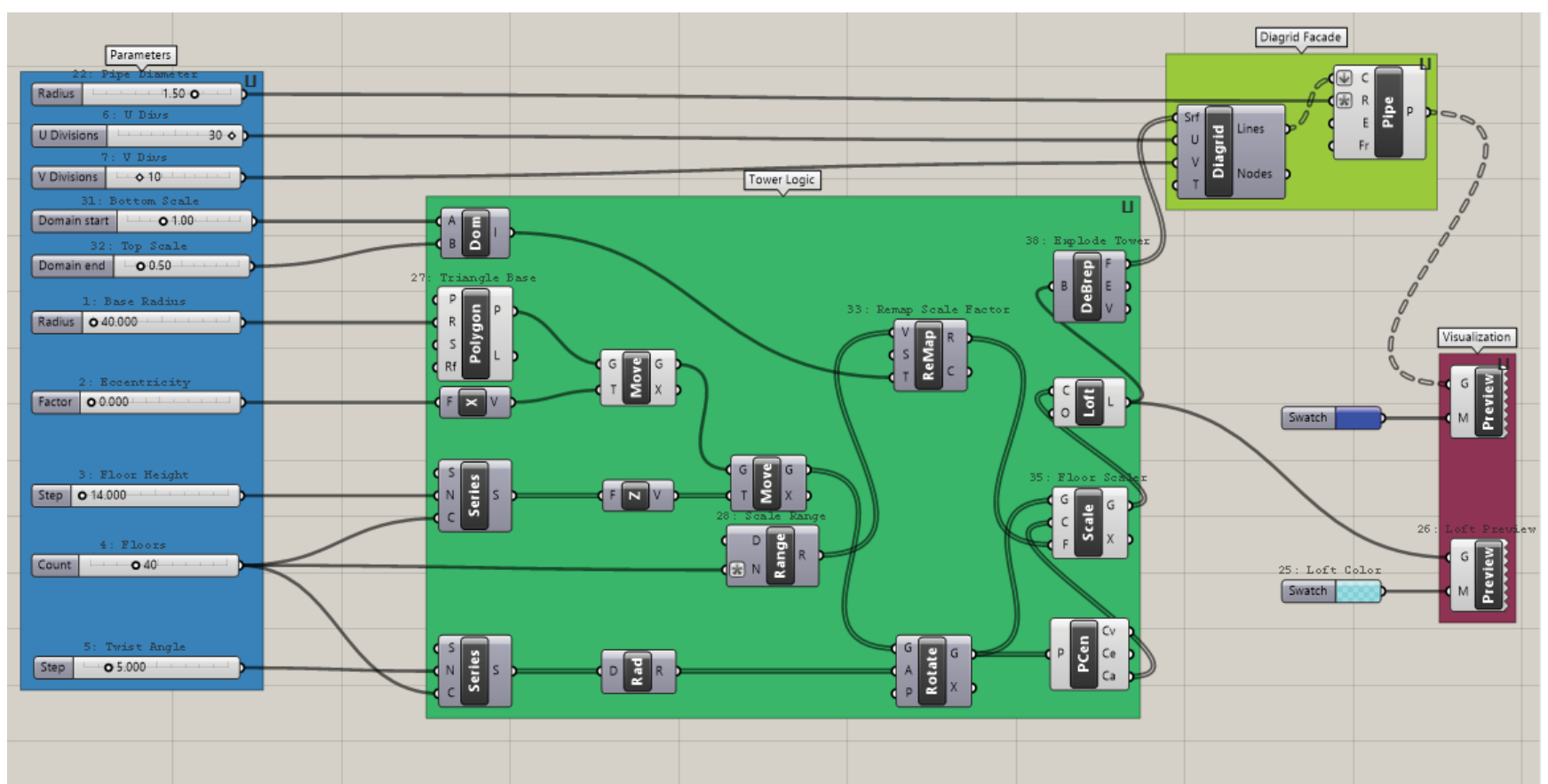


SCRIPT 1.1

RAVEN CHAT DOCUMENTATION



TOWER 2.0



SCRIPT 2.0

RAVEN CHAT DOCUMENTATION

Raven Chat

New Chat

Go Pro

Create a parametric, asymmetrical helix tower with a hexagonal facade.

My Thought Process for Crafting a Parametric, Asymmetrical Helix Tower with a Hexagonal Facade In Grasshopper >

Called ComponentSearch

Developing Asymmetrical Tower >

Called ComponentSearch

My Thought Process for Constructing the Elliptical Tower >

Called ComponentSearch

Mapping Graph Inputs >

Called ComponentSearch

My Thought Process as I Map the Graph >

Called ComponentSearch

Crafting a Parametric, Asymmetrical Helix Tower with a Hexagonal Facade >

Called EditScript

Undo

Correcting Component Selection >

Called ComponentSearch

My Thought Process on Resolving the Node 16 Issue and Refining Parameters >

Called EditScript

Undo

My Thought Process Unraveling a Node Instantiation Issue >

Called EditScript

Undo

Refining Facade Representation >

Called RelayoutGraph

Undo

Completing Tower Generation >

I have created a parametric, asymmetrical helix tower with a hexagonal facade.

Here is what I built:

Base Geometry: A circle offset from the origin (to create asymmetry when twisted).

Tower Setup: The circle is arrayed vertically and each curve is incrementally rotated around the origin axis.

Surface Generation: The rotated curves are lofted into a twisted, asymmetrical tower shape.

Hexagonal Facade: I used LunchBox.Hexagon Cells to map a structural hexagon pattern across the lofted surface and added a custom preview component.

You can tweak parameters like the twist angle, eccentricity, floor count, and grid division to change the shape and density of the facade.

Revise Floor Count slider to utilize whole numbers only.

Modifying Slider Settings >

Called EditScript

Undo

I have updated the "Floor Count" slider to use whole numbers only.

Add pipe geometry to the hexagonal facade grid with a slider to control the pipe diameter.

Adding Pipe Geometry >

Called ComponentSearch

Adding Pipe Geometry and Diameter Control >

Called EditScript

Undo

My Thought Process Regarding the Hexagonal Facade Pipe Geometry >

Called RelayoutGraph

Undo

I have added the pipe geometry to the hexagonal facade grid. A new slider named "Pipe Diameter" controls the pipe size, and the Custom Preview has been updated to display the structural pipes.

Let me know if you would like any further modifications!

Change hexagonal facade grid to a diagrid.

Exploring Diagrid Possibilities >

Called ComponentSearch

Replacing Hexagon with Diagrid >

Called EditScript

Undo

Integrating Diagrid Structure >

Called RelayoutGraph

Undo

I have updated the structure. The hexagonal cells have been swapped out for the LunchBox Diagrid Structure, providing a diagrid pattern across the lofted surface. The pipes now form the structural lines of this new diagrid facade.

Let me know if there are any other features you'd like to add!

how to adjust custom preview to turn off points and nodes

My Thought Process Regarding Custom Preview and Node Display >

To turn off the display of nodes (points), you generally need to disable the **Preview** of the component that is generating them. In your current script, the points are coming from the **Diagrid Structure** component (ID 24).

The **Custom Preview** component only displays the geometry connected to its input (which is currently your Pipes). It doesn't have a setting to hide other components' previews.

How to do it manually in Grasshopper:

Find the **Diagrid Structure** component.

Right-click the middle of the component and uncheck **Preview** (or select it and press Ctr1+Q).

This will hide the green/red points and wireframe lines from that specific component, leaving only the "Custom Preview" results visible.

If you'd like me to clean up the script or adjust any other logic, feel free to switch to **Edit mode**!

How do I modify the loft surface color?

Understanding the Goal and Identifying the Path Forward >

To modify the loft surface color, you need to add a second **Custom Preview** component specifically for that surface.

Currently, your **Custom Preview** (ID 20) is only showing the pipes. Here is how you can set it up:

Add a new **Custom Preview** component.

Add a new **Colour Swatch** component to choose your desired color.

Connect the **Loft output** (ID 17) to the *Geometry* input of the new preview.

Connect the new **Colour Swatch** to the *Material* input.

This will allow you to control the color of the tower's "skin" independently from the structural pipes.

If you'd like me to set this up for you, just switch to **Edit mode** and let me know!

CONT.

add a Custom Preview component to adjust the swatch color for the Loft output (ID17)

My Thought Process for Enhancing Lofted Surface Visualization >

Called EditScript

Undo

I have added a new **Colour Swatch** and **Custom Preview** for the lofted surface. You can now adjust the "Loft Color" independently from the diagrid pipes.

how do i group widgets?

My Thought Process on Grouping Widgets in Grasshopper >

In Grasshopper, you can group components by following these steps:

Select the components you want to group (hold Shift to select multiple).

Press **Ctr1 + G** on your keyboard (or go to Edit > Group).

To name the group: Right-click on the colored boundary of the group and type a name in the text box at the top of the menu.

To change the color: Right-click the group boundary and select a new color or style (like "Blob").

If you would like me to organize your script into labeled groups for you (e.g., "Inputs," "Tower Logic," "Facade"), just switch to **Edit mode** and let me know!

organize the script into labeled groups

My Thought Process for Script Organization >

Called EditGroups

Undo

I have organized your script into four labeled groups:

Parameters (Blue): All sliders and color swatches.

Tower Logic (Green): Helix geometry generation and rotation.

Diagrid Facade (Teal): The diagrid system and its piping logic.

Visualization (Maroon): Custom previews for the tower and facade.

Modify the script to utilize a triangle plane instead of a circle base plane and add a scalar control for each tower floor

My Thought Process for Modifying the Tower Script >

Called ComponentSearch

My Thought Process on Enhancing the Tower Geometry >

Called EditScript

Undo

My Thought Process on the Graph Mapper Issue and Related Adjustments >

Called EditScript

Undo

Addressing the Graph Mapper Configuration and System Organization >

Called EditScript

Undo

My Thought Process Regarding the Graph Mapper Issue >

Called EditGroups

Undo

My Thought Process: Transitioning from Graph Mapper to Python for Tower Scaling >

Called EditScript

Undo

My Thought Process on Resolving Script and Data Flow Issues >

Called EditScript

Undo

My Thought Process on Debugging the Geometry Generation >

Called EditScript

Undo

My Thought Process Regarding the Tower Geometry and Diagrid Error >

Called EditScript

Undo

My Thought Process on the Loft and Diagrid Integration >

Called ComponentSearch

Understanding the Diagrid Failure and My Path Forward >

Called EditScript

Undo

My Thought Process on Your Tower Revision >

Called EditGroups

Undo

My Thought Process on Adjusting Node Display >

To turn off the display of nodes (points) and wireframes, you must disable the **Preview** on the components that generate them. The **Custom Preview** component only *adds* geometry to the scene; it cannot hide geometry from other nodes.

Select the components producing the points (like **Diagrid Structure** or **Polygon Center**).

Press **Ctr1 + Q** to toggle their preview off.

Alternatively, right-click the center of the component and uncheck **Preview**.

I have highlighted the components in your script that are likely responsible for the visible points.

Called AddIdsToSelection

My Thought Process on Modifying Loft Surface Color and Script Organization >

Called EditGroups

Undo

My Thought Process on Refining the Script and Visual Display >

The tower now features a **triangle base** and a **tapering profile** controlled by "Bottom Scale" and "Top Scale" sliders. I have also organized the script into clean, labeled groups.

To hide the points and wireframes (the green/red lines and dots):

Select the **Diagrid Structure**, **Polygon Center**, and **Triangle Base** components (I have highlighted them for you).

Press **Ctr1 + Q** on your keyboard.

This will leave only the colored "Custom Preview" geometry visible. Is there anything else you'd like to adjust on the tower logic?

Tokens: 35

Type your message...

Edit

Fast

ADAM MORGAN

ELVTR | AI FOR ARCHITECTS - ASSIGNMENT 3